Artificial Intelligence in Mobile Forensics: A Survey of Current Status, a Use Case Analysis and AI Alignment Objectives

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Abstract

As the capabilities and utility of Artificial Intelligence and Machine Learning systems continue to improve, they are expected to have an increasingly powerful influence in the digital forensic investigation process. The concurrent proliferation of mobile devices and rapid increase of forensic value of related artifacts creates the requirement for a comprehensive review of the current status of artificial intelligence software usage and usefulness in Mobile Forensics. In this context, we conducted a survey to evaluate the characteristics and properties of AI functions in mobile forensic software from the practitioners' perspective and enhance understanding to the work in the field. In this study, we evaluated the performance of image categorization software in digital forensics using a variety of evaluation metrics including accuracy, precision, recall, and F1-score, as well as the confusion matrix. In this research we also identify and integrate theoretical principles to conceptualize an AI Alignment framework pertaining to Mobile Forensics and Digital Forensics in general, in order to accurately determine specific AI strategy objectives and potential solutions to the current technical and administrative landscape. We emphasized the importance of interpretability and transparency in AI systems and the need for a comprehensive approach to understanding the reasoning behind the software's decisions. Additionally, we highlighted the importance of robustness in image categorization software, as well as the consideration of AI governance and standardized procedures concepts. Our results show that the accuracy and robustness of the image categorization software have a significant impact on the outcome of legal cases and that the software should be designed with interpretability, transparency, and robustness in mind. Through the examination of the survey responses, the evaluation of the image categorization software and research literature, we explore existing and potential approaches to aligned Artificial Intelligence and analyze their contribution to the forensic examination of cases.

Keywords: Mobile forensics; Artificial intelligence; AI alignment; Forensic science; Digital forensics.

1. Introduction

The notion of decomposability of interpretability requires that inputs themselves be individually interpretable, disqualifying some models with highly engineered or anonymous features, while the notion of transparency might apply at the level of the learning algorithm itself, which relates to algorithmic transparency. Mordvintsev et al. (2015) attempt to explain what an image classification network has learned by altering the input through gradient descent to enhance the activations of certain nodes selected from the hidden layers.

In terms of standardization, the ISO/IEC 17020:2012 and the ISO/IEC 17025:2017 standards both include requirements for impartiality and freedom from bias. The importance and different aspects of bias are mentioned in “ISO Standards Addressing Issues of Bias and Impartiality in Forensic Work,” and a particular case can be made about the usage of AI. Shah et al.'s (2019) “On the Feasibility of Learning, Rather than Assuming, Compiled Human Biases for Reward Inference” argues that learning human values and biases at the same time, while

impossible in theory, is actually possible in practice.

Additionally, the idea has been put forward that we can build moral and interpretable AI by generalizing ethical rules from examples and giving justifications for actions in terms of deduction from ethical rules. In an important paper in the field (Atomium – European Institute for Science, Media and Democracy, 2018), the Atomium–EISMD initiative was presented, designed to lay the foundations for a "Good AI Society." This paper aims to provide a comprehensive overview of the main opportunities and risks of artificial intelligence (AI) for society, based on a review of the literature and expert opinions. It also proposes ethical principles to guide the development and adoption of AI, and concrete recommendations for policy makers, developers, and other stakeholders to safeguard the positive impact of AI on society. The concept has already affected policy measures in Europe and has considerable potential in the field of AI alignment. The importance of standards and policy making in correlation with digital forensic examinations using AI software are outlined in the respective section of the paper.

Following the above research efforts, the present study also aims to explore the current applications of AI in Mobile Forensics and the corresponding solutions it provides to the challenges that investigators face, as well as indicating particularly useful AI research topics that would benefit the field in the long term. In this context, a survey was conducted to address the current status of AI in the field of Mobile Forensics, the AI functions of forensic software, and their assessment by forensic examiners. In this research, we also focus on the performance evaluation of image categorization software in mobile forensics. We examine the use of various evaluation metrics, including accuracy, precision, recall, and F1-score, as well as the confusion matrix, to assess the performance of three prominent mobile forensics software products: Magnet Axiom, UFED Physical Analyzer, and Oxygen Forensic Detective. Additionally, we delve into the importance of interpretability, transparency, and robustness in AI systems, specifically in the context of image categorization software in mobile forensics. We also address the ethical and societal implications of the software, with the aim of developing image categorization software that is accurate, reliable, and ethically sound. Our study provides insight into the critical importance of evaluating the performance of image categorization software in mobile forensics and of taking an approach that considers interpretability, transparency, robustness, and ethical considerations.

Regarding the structure of the paper, Section 2 provides the methodology, procedure, and findings of the questionnaire survey related to the usage of Artificial Intelligence in Mobile Forensics. Section 3 examines the performance evaluation of image categorization software in Mobile Forensics, while Section 4 inspects the alignment landscape in Digital Forensics and the Mobile Forensics subfield and presents related transparency and robustness concepts. Finally, conclusions and recommendations for future work are presented in Section 5.

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